

Unit 9: Introduction to Calculus

Honors Precalculus

Student

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9.1 Finding Limits: Numerical and Graphical Approaches

Objectives

- Understand and interpret limit notation.
- Find a limit using a graph.
- Find a limit using a numerical table.
- Distinguish between a limit and a function value.

Limit Notation

A quantity L is the **limit** of $f(x)$ as x approaches a if, as input values of x approach a (but do not equal a), the corresponding output values get arbitrarily close to L .

Notation: $\lim_{x \rightarrow a} f(x) = L$

Read: “the limit of $f(x)$ as x approaches _____ equals _____.”

Key point: The limit describes behavior _____ a , not *at* a . The value $f(a)$ need not exist, and even if it does, it need not equal L .

Example 1

Identify a , $f(x)$, and L in the statement $\lim_{x \rightarrow 2} (3x + 5) = 11$.

One-Sided Limits and the Two-Sided Limit

Left-hand limit: $\lim_{x \rightarrow a^-} f(x) = L$ — x approaches a from the _____ ($x < a$).

Right-hand limit: $\lim_{x \rightarrow a^+} f(x) = L$ — x approaches a from the _____ ($x > a$).

Two-sided limit:

$\lim_{x \rightarrow a} f(x) = L$ if and only if $\lim_{x \rightarrow a^-} f(x) = \underline{\hspace{2cm}}$ and $\lim_{x \rightarrow a^+} f(x) = \underline{\hspace{2cm}}$

If the left and right limits are not equal, we say the limit _____ (DNE).

Example 2

A piecewise function has a graph where:

- As $x \rightarrow 2^-$, the graph approaches $y = 8$.
- As $x \rightarrow 2^+$, the graph approaches $y = 3$.
- The function has a filled dot at $(2, 3)$, so $f(2) = 3$.

Find $\lim_{x \rightarrow 2^-} f(x)$, $\lim_{x \rightarrow 2^+} f(x)$, $\lim_{x \rightarrow 2} f(x)$, and $f(2)$.

Example 3

A second piecewise function has:

- Left-hand limit at $x = 2$ equal to 8.
- Right-hand limit at $x = 2$ equal to 8.
- $f(2) = 4$ (filled dot below the approach value).

Find $\lim_{x \rightarrow 2} f(x)$ and $f(2)$. Are they equal?

Fact**Limit vs. Function Value**

The limit $\lim_{x \rightarrow a} f(x)$ and the function value $f(a)$ are **independent**: the limit may exist even if $f(a)$ is undefined, and the limit may differ from $f(a)$ even if both exist.

Note**Finding a Limit Numerically (Table Method):**

1. Choose x -values that approach a from both sides.
2. Compute $f(x)$ at each value.
3. If the outputs approach the same value L from both sides, then $\lim_{x \rightarrow a} f(x) = L$.

Note: The table gives evidence for the limit — it does not prove it. Algebraic methods (Section 9.2) provide proof.

Example 4

Use a table to estimate $\lim_{x \rightarrow 0} \frac{5 \sin x}{3x}$.

Example 5

Investigate $\lim_{x \rightarrow 0} 3 \sin\left(\frac{\pi}{x}\right)$. Does the limit exist?

Homework: OpenStax Section 12.1 Exercises (odds recommended)

9.2 Finding Limits: Properties of Limits**Objectives**

- Apply limit laws to evaluate limits algebraically.
- Use direct substitution for polynomial and power limits.
- Resolve the indeterminate $0/0$ form using factoring, simplification, or the conjugate technique.

Limit Laws

Let a , k , A , B be real numbers with $\lim_{x \rightarrow a} f(x) = A$ and $\lim_{x \rightarrow a} g(x) = B$.

Law	Formula	In words
Constant	$\lim_{x \rightarrow a} k = \underline{\hspace{2cm}}$	A constant stays itself
Constant multiple	$\lim_{x \rightarrow a} [k \cdot f] = \underline{\hspace{2cm}}$	Scalar passes through
Sum	$\lim_{x \rightarrow a} [f + g] = \underline{\hspace{2cm}}$	Limit distributes over +
Difference	$\lim_{x \rightarrow a} [f - g] = \underline{\hspace{2cm}}$	Limit distributes over −
Product	$\lim_{x \rightarrow a} [f \cdot g] = \underline{\hspace{2cm}}$	Limit distributes over $\times$
Quotient	$\lim_{x \rightarrow a} \frac{f}{g} = \underline{\hspace{2cm}}, B \neq 0$	Limit distributes over $\div$
Power	$\lim_{x \rightarrow a} [f]^n = \underline{\hspace{2cm}}$	Limit passes into exponent
n th root	$\lim_{x \rightarrow a} \sqrt[n]{f} = \underline{\hspace{2cm}}$	Limit passes into root

Polynomial rule: For any polynomial $p(x)$:

$$\lim_{x \rightarrow a} p(x) = \underline{\hspace{2cm}} \quad (\text{direct substitution})$$

Example 1

Evaluate $\lim_{x \rightarrow 3} (2x + 5)$.

Example 2

Evaluate $\lim_{x \rightarrow 5} (2x^3 - 3x + 1)$.

Example 3

Evaluate $\lim_{x \rightarrow 2} (3x + 1)^5$.

Note**The 0/0 Indeterminate Form**

If direct substitution gives $\frac{0}{0}$, the limit may still exist but **cannot be evaluated by substitution alone**. First simplify the expression algebraically, then take the limit.

Three techniques:

1. **Factoring:** Factor numerator (or denominator), then cancel common factors.
2. **Complex fractions:** Find a common denominator in the numerator, then simplify.
3. **Conjugate:** Multiply numerator and denominator by the conjugate to eliminate a radical.

Example 4

Evaluate $\lim_{x \rightarrow 2} \frac{x^2 - 6x + 8}{x - 2}$.

Example 5

Evaluate $\lim_{x \rightarrow 5} \frac{\frac{1}{x} - \frac{1}{5}}{x - 5}$.

Example 6

Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{25 - x} - 5}{x}$.

Example 7

Evaluate $\lim_{x \rightarrow 4} \frac{4 - x}{\sqrt{x} - 2}$.

Example 8

Evaluate $\lim_{x \rightarrow 7} \frac{|x - 7|}{x - 7}$.

Homework: OpenStax Section 12.2 Exercises (odds recommended)

9.3 Continuity**Objectives**

- State and apply the three conditions of continuity at a point.
- Identify removable and jump discontinuities.
- Determine whether a function is continuous at a given value.

Continuity at a Point

A function f is **continuous at** $x = a$ if **all three** of the following conditions hold:

1. _____
2. _____
3. _____

If any condition fails, the function is **discontinuous** at $x = a$. A function with **no** holes or breaks anywhere in its graph is called **continuous everywhere**.

Types of Discontinuity

Jump discontinuity: Both one-sided limits exist but are not equal:

$$\lim_{x \rightarrow a^-} f(x) \neq \lim_{x \rightarrow a^+} f(x)$$

The graph has a _____ at $x = a$. Cannot be fixed by redefining one point.

Removable discontinuity: The two-sided limit exists, but either $f(a)$ is _____ OR $f(a) \neq \lim_{x \rightarrow a} f(x)$. The graph has a _____ at $x = a$. Can be fixed by _____.

Example 1

Given $f(x) = \frac{x^2 - 25}{x - 5}$, determine whether f is continuous at $x = 5$. If not, classify the discontinuity.

Example 2

Is $f(x) = \begin{cases} 4x & x \leq 3 \\ 8 + x & x > 3 \end{cases}$ continuous at $x = 3$?

Example 3

A three-piece piecewise function is defined by:

$$f(x) = \begin{cases} x + 1 & x < 2 \\ 3 & x = 2 \\ x^2 - 1 & x > 2 \end{cases}$$

Is f continuous at $x = 2$?

Example 4

For $f(x) = \begin{cases} \sin x & x < 0 \\ x^3 & x > 0 \end{cases}$, determine continuity at $x = 0$.

Note**Functions That Are Continuous Everywhere:**

- Polynomial functions (e.g., $3x^4 - 2x + 1$)
- Exponential functions (e.g., e^x , 2^x)
- Sine and cosine functions ($\sin x$ and $\cos x$)

Functions Continuous on Their Domain:

- Logarithmic functions (defined for $x > 0$)
- Tangent function (undefined at $x = \frac{\pi}{2} + k\pi$)
- Rational functions (undefined where denominator = 0)

Homework: OpenStax Section 12.3 Exercises (odds recommended)

9.4 Derivatives

Objectives

- Compute the average rate of change between two points.
- Use the limit definition to find a derivative at a point.
- Write the equation of a tangent line.
- Find instantaneous velocity from a position function.

Average Rate of Change

The **average rate of change** (AROC) of f over the interval $[a, a + h]$ is:

$$\text{AROC} = \frac{f(a+h) - f(a)}{h}$$

This is also called the **difference quotient**.

Geometrically, it is the slope of the line connecting $(a, f(a))$ and $(a + h, f(a + h))$.

Example 1

Find the average rate of change of f between the points $(2, -6)$ and $(-1, 5)$.

The Derivative at a Point

The **derivative of f at $x = a$** , written $f'(a)$, is:

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

$f'(a)$ is the **instantaneous rate of change** of f at $x = a$.

Geometrically, $f'(a)$ is the slope of the _____ to the graph of f at the point $(a, f(a))$.

If the limit does not exist at $x = a$, the function is _____ there.

Note**Procedure for Finding $f'(a)$ from the Definition:**

1. Write $f(a+h)$ by substituting $a+h$ for x in $f(x)$.
2. Subtract $f(a)$ and simplify the numerator.
3. Divide by h and cancel the h factor.
4. Take the limit as $h \rightarrow 0$.

Example 2

Find $f'(a)$ for $f(x) = x^2 - 3x + 5$.

Example 3

Find $f'(a)$ for $f(x) = \frac{3+x}{2-x}$.

Example 4

Find $f'(36)$ for $f(x) = 4\sqrt{x}$.

Tangent Line Equation

The equation of the **tangent line** to $y = f(x)$ at $x = a$ is:

$$y = \underline{\hspace{2cm}}$$

This is point-slope form with slope $\underline{\hspace{2cm}}$ through the point $\underline{\hspace{2cm}}$.

Example 5

Find the equation of the tangent line to $f(x) = x^2 - 4x$ at $x = 3$.

Instantaneous Velocity

For a position function $s(t)$, the **instantaneous velocity** at time $t = a$ is:

$$s'(a) = \lim_{h \rightarrow 0} \underline{\hspace{2cm}}$$

- Positive $s'(a)$: moving in the $\underline{\hspace{2cm}}$ direction.
- Negative $s'(a)$: moving in the $\underline{\hspace{2cm}}$ direction.
- $\underline{\hspace{2cm}} = |s'(a)|$ (always non-negative).

Example 6

The height (in feet) of a ball at time t seconds is $s(t) = -16t^2 + 64t + 6$. Find the instantaneous velocity at $t = 1$ and at $t = 3$. Interpret each result.

Leibniz Notation and Non-Differentiability

Leibniz notation: The derivative is also written as $\frac{dy}{dx}$ or $\frac{df}{dx}$. The notation $\frac{dy}{dx}\big|_{x=a}$ means the derivative evaluated at $x = a$.

When is a function NOT differentiable at $x = a$?

- **Discontinuity:** f is not continuous at a .
- **Corner:** The graph has a sharp corner (e.g., $|x|$ at $x = 0$).
- **Cusp:** The graph has a sharp point with a vertical tangent.
- **Vertical tangent:** The tangent line is vertical (slope undefined).

Homework: OpenStax Section 12.4 Exercises (odds recommended)